

Restless Bandits | Whittle Index | Other Index options | Q-Learning

UQ ML Reading Seminar, Nov 6, 2025

Narrated by Yoni Nazarathy

Lagrange Index Policy for Restless Bandits with Average Reward

Konstantin Avrachenkov¹, Vivek S. Borkar² and Pratik Shah³

Abstract: We study the Lagrange Index Policy (LIP) for restless multi-armed bandits with long-run average reward. In particular, we compare the performance of LIP with the performance of the Whittle Index Policy (WIP), both heuristic policies known to be asymptotically optimal under certain natural conditions. Even though in most cases their performances are very similar, in the cases when WIP shows bad performance, LIP continues to perform very well. We then propose reinforcement learning algorithms, both tabular and NN-based, to obtain online learning schemes for LIP in the model-free setting. The proposed reinforcement learning schemes for LIP require significantly less memory than the analogous schemes for WIP. We calculate analytically the Lagrange index for the restart model, which applies to the optimal web crawling and the minimization of the weighted age of information. We also give a new proof of asymptotic optimality in case of homogeneous arms as the number of arms goes to infinity, based on exchangeability and de Finetti's theorem.

Key words: multi-armed bandits; restless bandits; Lagrange index; Whittle index; exchangeability

v2 [cs.LG] 26 Jun 2025

And two other papers

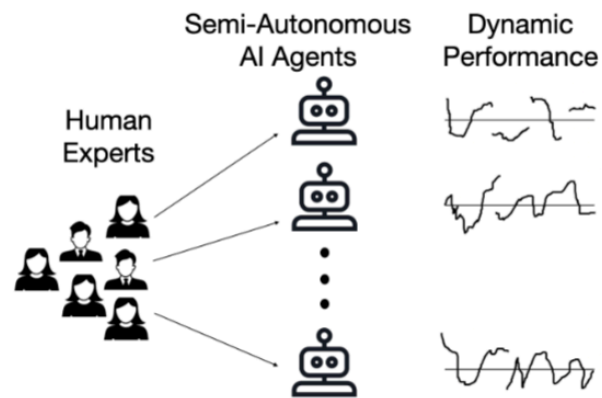


Figure 1: In a semi-autonomous systems framework, resources (e.g. human experts) are dynamically allocated to help semi-autonomous agents (e.g. AI agents) for improving performance.

Taken from UQ Control Theory (MATH) Course, 2014

Problem 5: Restless bandits

This problem deals with a type of MDP known as a *restless multi-armed bandit*. We don't describe the problem in generality here; we rather consider a useful common real life example: *Breastfeeding triplets*.

Consider a family that has just expanded from 2 (wife and husband) to 5 (wife, husband and triplets). Call the new babies, bandit 1, bandit 2 and bandit 3. The husband (who now became father), is rushing around all day, trying to make decisions on how and what to do. The wife (new mother), still just after birth, is sitting stationary, and most of the time breast-feeding... But as you might have figured out, she can't handle all three bandits at once. It is then up to the father to schedule the babies, moving them from their cot, to their mother's comforting nutrition, and back.

With an aim of being synchronised and efficient, every 30 minutes the father makes a transition, exchanging the bandits between the mother and their cots. So his decision at every such time-slot is: "where to place the bandits", with the obvious constraint of no more than 2 with mummy.

When a bandit is in his cot, refer to the bandit as *passive*. When the bandit is with mummy, refer to him as *active*. At this early point in their life, the bandits don't appear to be doing much. It is then sensible to model the *state* of each bandit by,

1 $\equiv$ deep sleep, 2 $\equiv$ awake and happy, 3 $\equiv$ screaming the house down.

As the new parents recently discovered, the bandits are not deterministic objects, but nevertheless, they seem to take well to breastfeeding and thus when active, a bandit changes state according to the following transition probability matrix:

$$P_a = \begin{bmatrix} 0.6 & 0.1 & 0.3 \\ 0.5 & 0.2 & 0.3 \\ 0.3 & 0.6 & 0.1 \end{bmatrix}.$$

On the other hand, a *passive* bandit changes state as follows:

$$P_p = \begin{bmatrix} 0.6 & 0.1 & 0.3 \\ 0.1 & 0.3 & 0.6 \\ 0.2 & 0.1 & 0.7 \end{bmatrix}.$$

In view of this data, the brave scheduling father has two main goals: (1) Minimise screaming. (2) Let the mother occasionally get some breastfeeding relief. For this he formulated the following cost structure per time interval:

cost = number of screaming bandits + number of feeding bandits.

For you: Formulate the problem as an MDP (state-space, action sets, transitions, rewards, etc...). Based on the problem data, think of some stationary Markov policy which you think is sensible. Specify your decision rule precisely. Run a long time horizon simulation (e.g. 10,000 time steps) to evaluate the average cost per step of your policy.

Some papers...

J. R. Statist. Soc. B (1979),
41, No. 2, pp. 148–177

Bandit Processes and Dynamic Allocation Indices

By J. C. GITTINS

Keble College, Oxford

[Read before the ROYAL STATISTICAL SOCIETY at a meeting organized by the RESEARCH SECTION on Wednesday, February 14th, 1979, the Chairman Professor J. F. C. KINGMAN in the Chair]

SUMMARY

The paper aims to give a unified account of the central concepts in recent work on bandit processes and dynamic allocation indices; to show how these reduce some previously intractable problems to the problem of calculating such indices; and to describe how these calculations may be carried out. Applications to stochastic scheduling, sequential clinical trials and a class of search problems are discussed.

Keywords: BANDIT PROCESSES; DYNAMIC ALLOCATION INDICES; TWO-ARMED BANDIT PROBLEM; MARKOV DECISION PROCESSES; OPTIMAL RESOURCE ALLOCATION; SEQUENTIAL RANDOM SAMPLING; CHEMICAL RESEARCH; CLINICAL TRIALS; SEARCH

1. INTRODUCTION

A scheduling problem

There are n jobs to be carried out by a single machine. The times taken to process the jobs are independent integer-valued random variables. The jobs must be processed one at a time. At the beginning of each time unit any job may be selected for processing, whether or not the job processed during the preceding time unit. A penalty or delay involved in switching from one job to another. The number of time units required to complete the processing of job i is t_i . The reward for processing job i at time t , is $p_i(t)$ ($i = 1, 2, \dots, n$; $t \in \mathbf{Z}$). The reward for not processing job i at time t , is 0 ($0 < a < 1$; $V_i > 0$; $i = 1, 2, \dots, n$). and there are no other jobs.

Bandit processes and dynamic allocation indices

JC Gittins - Journal of the Royal Statistical Society Series B ..., 1979 - academic.oup.com

... is applied at each stage, so that process time coincides with real time, is d ... bandit process is a bandit process for which, for some A, $R(x, l) = A$, $\forall x$. An arbitrary policy for a bandit process ...

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Dynamic ad allocation: Bandits with budgets

Restless Bandits: Activity Allocation in a Changing World

P. WHITTLE

Abstract

We consider a population of n projects which in general continue to evolve whether in operation or not (although by different rules). It is desired to choose the projects in operation at each instant of time so as to maximise the expected rate of reward, under a constraint upon the expected number of projects in operation. The Lagrange multiplier associated with this constraint defines an index which reduces to the Gittins index when projects not being operated are static. If one is constrained to operate m projects exactly then arguments are advanced to support the conjecture that, for m and n large in constant ratio, the policy of operating the m projects of largest current index is nearly optimal. The index is evaluated for some particular projects.

GITTINS INDEX; MULTI-ARMED BANDITS; SEQUENTIAL SCHEDULING; STIMULATING PRICES; INDEXABILITY

1. Introduction

The multi-armed bandit problem is a classic version of the problem of optimal allocation of activity under certainty. One can phrase it by saying that one has n projects, the state of project i being denoted by x_i (or by $x_i(t)$ if one wishes to emphasise its dependence on time, t). One can operate only one project at once: if one operates project i then one receives reward $g_i(x_i(t))$ in the time-interval $(t, t + 1)$ and the transition $x_i(t) \rightarrow x_i(t + 1)$ follows a Markov

Restless bandits: Activity allocation in a changing world

P Whittle - Journal of applied probability, 1988 - cambridge.org

We consider a population of n projects which in general continue to evolve whether in operation or not (although by different rules). It is desired to choose the projects in operation at ...

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ON AN INDEX POLICY FOR RESTLESS BANDITS

RICHARD R. WEBER,* *University of Cambridge*
GIDEON WEISS,** *Georgia Institute of Technology*

Abstract

We investigate the optimal allocation of effort to a collection of n projects. The projects are 'restless' in that the state of a project evolves in time, whether or not it is allocated effort. The evolution of the state of each project follows a Markov rule, but transitions and rewards depend on whether or not the project receives effort. The objective is to maximize the expected time-average reward under a constraint that exactly m of the n projects receive effort at any one time. We show that as m and n tend to ∞ with m/n fixed, the per-project reward of the optimal policy is asymptotically the same as that achieved by a policy which operates under the relaxed constraint that an average of m projects be active. The relaxed constraint was considered by Whittle (1988) who described how to use a Lagrangian multiplier approach to assign indices to the projects. He conjectured that the policy of allocating effort to the m projects of greatest index is asymptotically optimal as m and n tend to ∞ . We show that the conjecture is true if the differential equation describing the fluid approximation to the index policy has a globally stable equilibrium point. This need not be the case, and we present an example for which the index policy is not asymptotically optimal. However, numerical work suggests that such counterexamples are extremely rare and that the size of the suboptimality which one might expect is minuscule.

FLUID APPROXIMATIONS; GITTINS INDEX; LARGE DEVIATION THEORY; MULTI-ARMED BANDIT PROBLEM; STOCHASTIC SCHEDULING

1. Restless bandits

Whittle (1988) has recently studied the multi-armed bandit problem. The state of project i at time t being denoted $x_i(t)$, project i is operated then a reward

On an index policy for restless bandits

RR Weber, G Weiss - Journal of applied probability, 1990 - cambridge.org

... for which the index policy is not **asymptotically** optimal. However... Whittle has studied a variation in which two generalizations ... Whittle gave several examples of problems which are nicely ...

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[52] Weber, R. R. and Weiss, G., 1990. On an index policy for restless bandits. Journal of Applied Probability, 27(3), 637-648.

THE COMPLEXITY OF OPTIMAL QUEUEING NETWORK CONTROL

CHRISTOS H. PAPADIMITRIOU AND JOHN N. TSITSIKLIS

We show that several well-known optimization problems related to the optimal control of queues are *provably intractable*—independently of any unproven conjecture such as $P \neq NP$. In particular, we show that several versions of the problem of optimally controlling a simple network of queues with simple arrival and service distributions and multiple customer classes is complete for exponential time. This is perhaps the first such intractability result for a well-known optimization problem. We also show that the restless bandit problem (the generalization of the multi-armed bandit problem to the case in which the unselected processes are not quiescent) is complete for polynomial space.

1. Introduction. The optimal control of a network of queues is a well-known, much studied, and notoriously difficult problem. We are given several servers, a set of customer classes, and class-dependent probability distributions for the service times. For each customer class, there is only one server that can serve customers of that class, but the same server may be eligible for several classes. The class of a customer can change at each service completion time; for some customer classes, the new class is under our control; for others, the change is probabilistic. We restrict ourselves to closed networks in which there is a finite number of customers that never leave the system, and no external arrivals. The throughput of a class of customers is defined as the steady-state average number of service completions for that class per unit time; our performance measure will be a weighted sum of throughputs of the different classes. Operating the network amounts to choosing the class of a customer whose service has just been completed (routing) and choosing at each server which customer to serve next, out of all eligible customers (sequencing). The problem is to come up with a routing and sequencing strategy—with decisions presumably based on the load of the queues—so as to optimize weighted throughput.

The complexity of optimal queueing network control

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CH Papadimitriou, JN Tsitsiklis

Mathematics of Operations Research, 1999 · pubsonline.informs.org

We show that several well-known optimization problems related to the optimal control of queues are *provably intractable*—independently of any unproven conjecture such as $P \neq NP$. In particular, we show that several versions of the problem of optimally controlling a simple network of queues with simple arrival and service distributions and multiple customer classes is complete for exponential time. This is perhaps the first such intractability result for a well-known optimization problem. We also show that the restless

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ASYMPTOTICALLY OPTIMAL PRIORITY POLICIES FOR INDEXABLE AND NONINDEXABLE RESTLESS BANDITS

BY I. M. VERLOOP

CNRS, IRIT, Université de Toulouse and INP

We study the asymptotic optimal control of multi-class restless bandits. A restless bandit is a controllable stochastic process whose state evolution depends on whether or not the bandit is made active. Since finding the optimal control is typically intractable, we propose a class of priority policies that are proved to be asymptotically optimal under a global attractor property and a technical condition. We consider both a fixed population of bandits as well as a dynamic population where bandits can depart and arrive. As an example of a dynamic population of bandits, we analyze a multi-class $M/M/S + M$ queue for which we show asymptotic optimality of an index policy.

We combine fluid-scaling techniques with linear programming results to prove that when bandits are indexable, Whittle's index policy is included in our class of priority policies. We thereby generalize a result of Weber and Weiss [*J. Appl. Probab.* **27** (1990) 637–648] about asymptotic optimality of Whittle's index policy to settings with (i) several classes of bandits, (ii) arrivals of new bandits and (iii) multiple actions.

Indexability of the bandits is not required for our results to hold. For non-indexable bandits, we describe how to select priority policies from the class of asymptotically optimal policies and present numerical evidence that, outside the asymptotic regime, the performance of our proposed priority policies is nearly optimal.

Asymptotically optimal priority policies for indexable and nonindexable restless bandits

[IM Verloop](#)

2016 · [projecteuclid.org](#)

Abstract

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Towards Q-learning the Whittle Index for Restless Bandits

Jing Fu¹, Yoni Nazarathy², Sarat Moka² and Peter G. Taylor¹

Abstract—We consider the multi-armed restless bandit problem (RMABP) with an infinite horizon average cost objective. Each *arm* of the RMABP is associated with a Markov process that operates in two modes: *active* and *passive*. At each time slot a controller needs to designate a subset of the arms to be active, of which the associated processes will evolve differently from the passive case. Treated as an optimal control problem, the optimal solution of the RMABP is known to be computationally intractable. In many cases, the *Whittle index policy* achieves near optimal performance and can be tractably found. Nevertheless, computation of the *Whittle indices* requires knowledge of the transition matrices of the underlying processes, which are sometimes hidden from decision makers. In this paper, we take first steps towards a tractable and efficient reinforcement learning algorithm for controlling such a system. We setup parallel Q-learning recursions, with each recursion mapping to individual possible values of the Whittle index. We then update these recursions as we control the system, learning an approximation of the Whittle index as time evolves. Tested on several examples, our control outperforms naive priority allocations and nears the performance of the fully-informed Whittle index policy.

I. INTRODUCTION

The multi-armed bandit problem (MABP) is a decision making problem that sequentially *activates* one out of I parallel Markov processes and leaves the remaining $I - 1$ *passive*: the passive processes will be frozen (with no state

The Whittle index policy, similar to Gittins index policy, always prioritizes the processes with the highest state-dependent indices, referred to as the *Whittle indices*, which are calculated by solving *sub-problems* with remarkably reduced state spaces. The (R)MABP has been widely applied in resource allocation problems such as job scheduling in cloud computing [6], channel detection in communications [7], the health care system [8], and dynamic posted pricing [9]. However, calculation or approximation of Whittle indices generally requires full knowledge of the transition matrices of all the Markov processes. In practice, this is rarely known by system controllers.

Problems based on Markov decision process (MDP) with hidden transition matrices have been analyzed through Q-learning (or deep Q-learning) for decades [10]–[13]. Although (deep) Q-learning has been demonstrated to be powerful in approximating value functions, the convergence time to an optimal solution for a problem with a large state space is still an open question. Hence Q-learning of RMABP using a straightforward approach is not applicable.

Borkar and Chadha [14] adapted a Q-learning approach (temporal difference method), which was proposed in [12], to approximate the Whittle index in a RMABP with continuous state space.

Towards q-learning the whittle index for restless bandits

..., Y. Nazarathy, S. Moka, PG Taylor - 2019 Australian & New ..., 2019 - ieeexplore.ieee.org

We consider the multi-armed restless bandit problem (RMABP) with an infinite horizon average cost objective. Each arm of the RMABP is associated with a Markov process that ...

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Whittle index based Q-learning for restless bandits with average reward[☆]

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ABSTRACT

A novel reinforcement learning algorithm is introduced for multiarmed restless bandits with average reward, using the paradigms of Q-learning and Whittle index. Specifically, we leverage the structure of the Whittle index policy to reduce the search space of Q-learning, resulting in major computational gains. Rigorous convergence analysis is provided, supported by numerical experiments. The numerical experiments show excellent empirical performance of the proposed scheme.

[HTML] Whittle index based Q-learning for restless bandits with average reward

[KE Avrachenkov](#), [VS Borkar](#)

Automatica, 2022 · Elsevier

Abstract

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Tabular and Deep Learning for the Whittle Index

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VIVEK BORKAR, Indian Institute of Technology, Mumbai, India

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KONSTANTIN AVRACHENKOV, Inria, Sophia Antipolis, France

The Whittle index policy is a heuristic that has shown remarkably good performance (with guaranteed asymptotic optimality) when applied to the class of problems known as Restless Multi-Armed Bandit Problems (RMABPs). In this article, we present QWI and QWINN, two reinforcement learning algorithms, respectively tabular and deep, to learn the Whittle index for the total discounted criterion. The key feature is the use of two time-scales, a faster one to update the state-action Q -values, and a relatively slower one to update the Whittle indices. In our main theoretical result, we show that QWI, which is a tabular implementation, converges to the real Whittle indices. We then present QWINN, an adaptation of QWI algorithm using neural networks to compute the Q -values on the faster time-scale, which is able to extrapolate information from one state to another and scales naturally to large state-space environments. For QWINN, we show that all local minima of the Bellman error are locally stable equilibria, which is the first result of its kind for DQN-based schemes. Numerical computations show that QWI and QWINN converge faster than the standard Q -learning algorithm, neural-network based approximate Q -learning, and other state-of-the-art algorithms.

CCS Concepts: • **Computing methodologies** → **Sequential decision making**;

Additional Key Words and Phrases: Machine learning, reinforcement learning, whittle index, Markov decision problem, multi-armed restless bandit

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Tabular and deep learning for the whittle index

F Robledo Relaño, V Borkar, U Ayesta, K Avrachenkov

ACM Transactions on Modeling and Performance Evaluation of Computing Systems, 2024 - dl.acm.org

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Testing Indexability and Computing Whittle and Gittins Index in Subcubic Time

Nicolas Gast, Bruno Gaujal and ✉ Kimang Khun

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Abstract

Whittle index is a generalization of Gittins index that provides very efficient allocation rules for restless multi-armed bandits. In this work, we develop an algorithm to test the indexability and compute the Whittle indices of any finite-state restless bandit arm. This algorithm works in the discounted and non-discounted cases, and can compute Gittins index. Our algorithm builds on three tools: (1) a careful characterization of Whittle index that allows one to compute recursively the k th smallest index from the $(k - 1)$ th smallest and to test indexability, (2) the use of the Sherman-Morrison formula to make this recursive computation efficient, and (3) a sporadic use of the fastest matrix inversion and multiplication methods to obtain a subcubic complexity. We show that an efficient use of the Sherman-Morrison formula leads to an algorithm that computes Whittle index in $(2/3)n^3 + o(n^3)$ arithmetic operations where n is the number of states of the arm. The careful use of fast matrix multiplication leads to the first subcubic algorithm to compute Whittle or Gittins index: By using the current fastest matrix multiplication, the theoretical complexity of our algorithm is $O(n^{2.5286})$. We also develop an efficient implementation of our algorithm that can compute indices of Markov chains with several thousands of states in less than a few seconds.

Keywords: Whittle Index, Gittins Index, Restless Bandit, Multi-armed Bandit, Sherman-Morrison, Markov Decision Process, Fast Matrix Multiplication

Testing indexability and computing Whittle and Gittins index in subcubic time

[N Gast](#), [B Gaujal](#), [K Khun](#)

Mathematical Methods of Operations Research, 2023 · Springer

Abstract

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[25] Gast, N., Gaujal, B. and Khun, K., 2023. Testing indexability and computing Whittle and Gittins index in subcubic time. Mathematical Methods of Operations Research, 97(3), 391-436.

Whittle Index...

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Problems based on Markov decision process (MDP) with hidden transition matrices have been analyzed through Q-learning (or deep Q-learning) for decades [10]–[13]. Although (deep) Q-learning has been demonstrated to be powerful in approximating value functions, the convergence time to an optimal solution for a problem with a large state space is still an open question. Hence Q-learning of RMABP using a straightforward approach is not applicable.

Borkar and Chadha [14] adapted a Q-learning approach (temporal difference method), which was proposed in [12], to approximate the Whittle index in a RMABP with continuous state space. Given a fixed Whittle index, they proved the convergence of the differential cost function (or bias func-

Consider I parallel *projects*, each of which is associated with a discrete-time Markov process, denoted by $\{X_i(t), t \in \mathbb{N}_0\}$, $i \in [I]$. A project can behave in two modes, *active* and *passive*, with potentially different transition matrices. At time t , we can choose to activate a project $i \in [I]$ or not, indicated by an *action variable* $A_i(t) \in \{0, 1\}$: if $A_i(t) = 1$, the project is active; otherwise, passive.

In this paper, all the I projects are identical in that they have the same state space, $\mathcal{X}$, and the same state transition matrices, $\mathcal{P}^1, \mathcal{P}^0 \in [0, 1]^{| \mathcal{X} | \times | \mathcal{X} |}$ for active and passive modes, respectively. The state space $\mathcal{X}$ is assumed to be finite with states labeled by $x = 1, 2, \dots, | \mathcal{X} |$. Let $P_{x,x'}^1$ and $P_{x,x'}^0$ ($x, x' \in \mathcal{X}$) represent the entries of $\mathcal{P}^1$ and $\mathcal{P}^0$ at row x and column x' , respectively. Let $\mathbf{X}(t) := (X_i(t) : i \in [I])$ and $\mathbf{A}(t) := (A_i(t) : i \in [I])$.

The process of the entire system, $\{\mathbf{X}(t), \mathbf{A}(t)\}_{t=0}^\infty$, acts as a *Markov decision process* (MDP) with $\mathcal{X}^I$ and $\{0, 1\}^I$ the state and action spaces, respectively. The rule that decides the probability of $\mathbf{A}(t)$ taking values in $\{0, 1\}^I$ is referred to as a *policy*. In particular, we limit this work to stationary policies; that is, $A_i(t) := a_i(\mathbf{X}(t))$, which is a function of state vector, and we refer to $a_i(\cdot)$ as the action variable for project $i \in [I]$. In this context, all action variables $a_i(\mathbf{x})$ for $i \in [I]$ and $\mathbf{x} \in \mathcal{X}^I$ determine a policy, denoted by ϕ . By slightly abusing notation, we rewrite the action and state variables as $a_i^\phi(\cdot)$ and $\mathbf{X}^\phi(t)$ to indicate their dependencies on policy ϕ . Let $\mathbf{a}^\phi(\mathbf{x}) := (a_i^\phi(\mathbf{x}) : i \in [I])$ for any $\mathbf{x} \in \mathcal{X}^I$, and Φ represent the set of all such policies ϕ .

By taking an action $a_i^\phi(\mathbf{x}) = a$ in state $\mathbf{X}^\phi(t) = \mathbf{x}$ ($\mathbf{x} \in \mathcal{X}^I$, $a \in \{0, 1\}$) at time t , an expected reward $R(x_i, a)$, which is bounded, for project $i \in [I]$ is generated. The total reward of the system generated at each time slot is the sum of the rewards for each project. At each time t , we are required to activate exactly $K \leq I$ of the I projects, so that

$$\sum_{i \in [I]} a_i^\phi(\mathbf{X}^\phi(t)) = K, \quad (1)$$

and we aim to maximize the long-run average reward of the entire system. To this end, for initial state $\mathbf{x}(0)$, our optimization problem can be written as

$$\max_{\phi \in \Phi} \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \sum_{i \in [I]} \mathbb{E}[R(X_i^\phi(t), a_i^\phi(\mathbf{X}^\phi(t)))], \quad (2)$$

subject to (1).

This problem is a standard *Restless Multi-Armed Bandit Problem* (RMABP), of which state space size is increasing exponentially with the number of projects I . The RMABP has been proved to be PSPACE-hard [3]. Thus, we resort

IV. WHITTLE INDICES

The Whittle indices are real values assigned to each state in $\mathcal{X}$ and are computed offline by optimizing all the I project processes independently. This idea remarkably reduces the computational complexity of solving a problem over state space $\mathcal{X}^I$ to that over $\mathcal{X}$.

The Whittle index policy has later been proved to be asymptotically optimal under certain conditions [4]. Also, the asymptotic optimality of the Whittle index policy or Whittle-like index policies has been proved or discussed in recent work [6], [15], [16]. In particular, for value functions that solve

$$g + V(x, \lambda) = \max \left\{ R(x, 1) - \lambda + \sum_{x' \in \mathcal{X}} P_{x,x'}^1 V(x', \lambda), \right. \\ \left. R(x, 0) + \sum_{x' \in \mathcal{X}} P_{x,x'}^0 V(x', \lambda) \right\} \quad (3)$$

with $x \in \mathcal{X}$ and $g, \lambda \in \mathbb{R}$, the Whittle index (if it exists) of a state $x \in \mathcal{X}$ is a specific value $\lambda(x) \in \mathbb{R}$, such that,

$$R(x, 1) - \lambda(x) + \sum_{x' \in \mathcal{X}} P_{x,x'}^1 V(x', \lambda(x)) \\ = R(x, 0) + \sum_{x' \in \mathcal{X}} P_{x,x'}^0 V(x', \lambda(x)). \quad (4)$$

If the $\lambda(x)$ in (3) is not unique, then we can take any one.

We briefly explain the notation in equation (3). For a given $\lambda \in \mathbb{R}$, consider a MDP that evolves in the same way as a project defined in Section II, except that this MDP gains reward $R(x, 1) - \lambda$ if it is active in state $x \in \mathcal{X}$. We refer to such a MDP as a λ -project. For a given $\lambda \in \mathbb{R}$, the $g \in \mathbb{R}$ in (3) is a given parameter that is equal to the maximized long-run average revenue of this λ -project. An optimal policy, which maximizes the right hand side of (3) for all $x \in \mathcal{X}$, will maximize the long-run average revenue of the λ -project [17]. The Whittle index (if it exists) for state $x \in \mathcal{X}$ is the specific value of λ such that active and passive actions in state x make no difference in terms of the average revenue of the λ -project, as described in (4).

V. Q-LEARNING FOR WHITTLE INDICES

By time t , we can retrieve information of the stochastic process with unknown transition matrices from the history of states and actions to solve the optimization problem described by (2) and (1). Let $\mathcal{H}(t) := \{\mathbf{X}^\phi(\tau), \mathbf{A}^\phi(\tau), \tau \in [t]\}$ represent the history of states and actions of the system up to time $t \in \mathbb{N}_0$.

A. Q-Learning

We adapt the *Q-learning* technique to our problem. The Q-learning technique often applies to a MDP without specific knowledge of its transition rules, and is used to learn the value functions according to observations. For instance, consider the value functions described in (3), and for $x \in \mathcal{X}$, $a \in \{0, 1\}$, $\lambda \in \mathbb{R}$, define the *Q function* as,

$$g + Q^\lambda(x, a) = R(x, a) - \lambda a + \sum_{x' \in \mathcal{X}} P_{x,x'}^a V(x', \lambda), \quad (5)$$

and $V(x, \lambda) = \max_{a \in \{0,1\}} Q^\lambda(x, a)$, where λ is considered as a given constant.

Also, define the *estimated Q function* as $\hat{Q}_t^\lambda(x, a)$ for $x \in \mathcal{X}$ and $a \in \{0, 1\}$, which is the estimated value of $Q^\lambda(x, a)$ according to the observed history $\mathcal{H}_t$. Since all project processes $i \in [I]$ are stochastically identical, we start our discussion with any of them and write the state variable as $X^\phi(t)$ and $A^\phi(t)$ by omitting the subscript i .

Q-learning is an iterative process that starts out with a set of $\hat{Q}_0^\lambda(x, a)$ for all $x \in \mathcal{X}$ and $a \in \{0, 1\}$ and updates them. The process is operated in a way that optimizes the estimated Q function. For $t \in \mathbb{N}_+$, it takes the action $A^\phi(t) = \arg \max_{a \in \{0,1\}} \{\hat{Q}_t^\lambda(X^\phi(t), a)\}$ and updates according to the iteration in (6), where β and $\{\alpha_t\}_{t=0}^\infty$ are hyper-parameters. Given the observation $X^\phi(t+1)$ at time $t+1$ after taking the action $A^\phi(t)$, equation (6) is a standard method in Q-learning that updates the estimated Q functions. Note that in (6), the transition matrices are not required. However it increments slowly; for $x \in \mathcal{X}$ and $a \in \{0, 1\}$, if $x \neq X^\phi(t)$ or $a \neq A^\phi(t)$, the estimated Q function $\hat{Q}_{t+1}^\lambda(x, a) = \hat{Q}_t^\lambda(x, a)$.

$$g + V(x, \lambda) = \max \left\{ R(x, 1) - \lambda + \sum_{x' \in \mathcal{X}} P_{x,x'}^1 V(x', \lambda), \right. \\ \left. R(x, 0) + \sum_{x' \in \mathcal{X}} P_{x,x'}^0 V(x', \lambda) \right\} \quad (3)$$

$$\hat{Q}_{t+1}^\lambda(X^\phi(t), A^\phi(t)) = (1 - \alpha_t) \hat{Q}_t^\lambda(X^\phi(t), A^\phi(t)) + \alpha_t \left(R(X^\phi(t), A^\phi(t)) - \lambda A^\phi(t) + \beta \max_{a \in \{0,1\}} \hat{Q}_t^\lambda(X^\phi(t+1), a) \right) \quad (6)$$

B. Q-Learning the Whittle Indices

Recall the definition of the Whittle indices $\lambda(x)$ ($x \in \mathcal{X}$) and the Q function in (4) and (5). We obtain

$$Q^{\lambda(x)}(x, 1) - Q^{\lambda(x)}(x, 0) = 0. \quad (7)$$

For state $x \in \mathcal{X}$, consider a set of candidate values of $\lambda(x)$, denoted by Λ , and, for $\lambda \in \Lambda$, $a \in \{0, 1\}$, $t \in \mathbb{N}_+$, define $Q(\lambda, x, a) = Q^\lambda(x, a)$ and $\hat{Q}_t(\lambda, x, a) = \hat{Q}_t(\lambda, x, a)$. In this context, λ becomes part of the state in a Q function, and we approximate the Whittle index $\lambda(x)$ ($x \in \mathcal{X}$) by optimizing

$$\min_{\lambda \in \Lambda} |\hat{Q}_t(\lambda, x, 1) - \hat{Q}_t(\lambda, x, 0)| \quad (8)$$

for each $t \in \mathbb{N}_+$. Define $\hat{\lambda}_t(x)$ as the *estimated Whittle index* for state $x \in \mathcal{X}$ at time t . The $\hat{\lambda}_t(x)$ is updated to be the λ minimizing (8).

On the other hand, these estimated Whittle indices $\hat{\lambda}_t(\cdot)$ will indicate the priorities of projects at time t : the K projects with highest $\hat{\lambda}_t(X_i^\phi(t))$ will be activated ($A_i^\phi(t) = 1$), and all the others will be passive. The estimated Q function $\hat{Q}_{t+1}(\cdot, \cdot, \cdot)$ and the estimated Whittle indices $\hat{\lambda}_{t+1}(\cdot)$ are then updated according to new observations $\mathbf{X}^\phi(t+1)$ and

updated $\hat{Q}_{t+1}(\cdot, \cdot, \cdot)$, by invoking (8) and (9).

```

Input :  $\mathbf{x}_0, \mathbf{x}_1 \in \mathcal{X}^I$ ,  $Q_0 \in \mathbb{R}^{\Lambda \times \mathcal{X} \times \{0,1\}}$ ,  $\hat{\lambda}_0 \in \Lambda^{\mathcal{X}}$ ,  $\mathbf{a}_0 \in \{0,1\}^I$  and  $t \in \mathbb{N}_+$ 
        /*  $\mathbf{x}_1$  is the newly observed state vector for a tested time slot  $t$ . */
        /*  $\mathbf{x}_0, Q_0, \hat{\lambda}_0, \mathbf{a}_0$  are the state vector, estimated Q function and Whittle index */
        /* and action vector for the previous time slot, respectively. */
Output:  $Q_1 \in \mathbb{R}^{\Lambda \times \mathcal{X} \times \{0,1\}}$ ,  $\hat{\lambda}_1 \in \Lambda^{\mathcal{X}}$  and  $\mathbf{a}_1 \in \{0,1\}^I$ 
        /*  $Q_1, \hat{\lambda}_1, \mathbf{a}_1$  are the updated Q function, the resulting estimated Whittle */
        /* index and the resulting action vector for the tested time slot. */

1 Function ( $Q_1, \hat{\lambda}_1, \mathbf{a}_1$ )  $\leftarrow$  QWICPolicy( $\mathbf{x}_0, Q_0, \hat{\lambda}_0, \mathbf{a}_0, \mathbf{x}_1$ ):
2    $Q_1(\lambda, x, a) \leftarrow Q_0(\lambda, x, a)$  for all  $\lambda \in \Lambda$ ,  $x \in \mathcal{X}$  and  $a \in \{0,1\}$ 
3   for  $i \in [I]$  do
4      $Q_1(\hat{\lambda}_0(x_{0,i}), x_{0,i}, a_{0,i}) \leftarrow$ 
        $(1 - \alpha_t) \hat{Q}_0(\hat{\lambda}_0(x_{0,i}), x_{0,i}, a_{0,i}) + \alpha_t \left( R(x_{0,i}, a_{0,i}) - \hat{\lambda}(x_{0,i}) a_{0,i} + \beta \max_{a \in \{0,1\}} \hat{Q}_0(\hat{\lambda}_0(x_{0,i}), x_{1,i}, a) \right)$ 
       /* Update  $Q_1$  by plugging parameters in (9). */
5   end
6   for  $x \in \mathcal{X}$  do
7      $\hat{\lambda}_1(x) \leftarrow \arg \min_{\lambda \in \Lambda} |Q_1(\lambda, x, 1) - Q_1(\lambda, x, 0)|$ 
8   end
9    $\mathcal{J} \leftarrow$  a subset of  $[I]$ : if  $i \in \mathcal{J}$ , then  $\hat{\lambda}(x_{1,i})$  is one of the  $K$  largest  $\hat{\lambda}(x_{1,i'})$  among all  $i' \in [I]$ 
       /* Guarantee  $|\mathcal{J}| = K$  by breaking tie cases randomly. */

10  for  $i \in [I]$  do
11    if  $i \in \mathcal{J}$  then
12       $a_{1,i} \leftarrow 1$ 
13    else
14       $a_{1,i} \leftarrow 0$ 
15    end
16  end
17  With probability  $\gamma_t$ , randomizes the elements of  $\mathbf{a}_1$  in place, and, for all  $x \in \mathcal{X}$ ,  $\hat{\lambda}_1(x) \leftarrow$  a uniformly randomly selected
    value from  $\Lambda$ .
18  return ( $Q_1, \hat{\lambda}_1, \mathbf{a}_1$ )
19 End

```

Algorithm 1: Pseudo-code for QWIC.

$$\begin{aligned} \hat{Q}_{t+1}(\hat{\lambda}(X^\phi(t)), X^\phi(t), A^\phi(t)) &= (1 - \alpha_t) \hat{Q}_t(\hat{\lambda}(X^\phi(t)), X^\phi(t), A^\phi(t)) \\ &+ \alpha_t \left(R(X^\phi(t), A^\phi(t)) - \hat{\lambda}(X^\phi(t)) A^\phi(t) + \beta \max_{a \in \{0,1\}} \hat{Q}_t(\hat{\lambda}(X^\phi(t)), X^\phi(t+1), a) \right) \end{aligned} \quad (9)$$

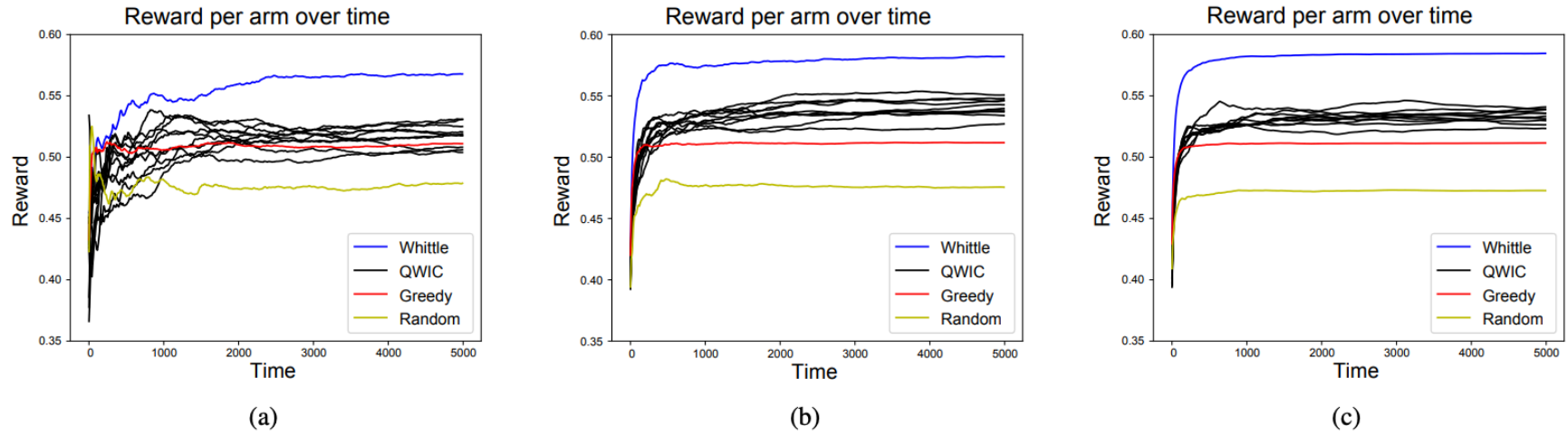


Fig. 3. Accumulated average reward for the Mentoring Instruction system: (a) $K = 1, I = 5$; (b) $K = 10, I = 50$; and (c) $K = 100, I = 500$.

reward indicated by the study level:

$$R(x, a) = c\sqrt{x} \text{ with } c = 1/\sqrt{10} \text{ a}$$

$$\mathcal{P}^1, \mathcal{P}^0 = \begin{bmatrix} 0.3 & 0.7 & & & \\ 0.3 & 0 & \ddots & & \\ & 0.3 & \ddots & 0.7 & \\ & & \ddots & 0 & 0.7 \\ & & & 0.3 & 0.7 \end{bmatrix}, \begin{bmatrix} 0.7 & 0.3 & & & \\ 0.7 & 0 & \ddots & & \\ & 0.7 & \ddots & 0.3 & \\ & & \ddots & 0 & 0.3 \\ & & & 0.7 & 0.3 \end{bmatrix},$$



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Whittle index based Q-learning for restless bandits with average reward[☆]

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ABSTRACT

Specifically, we consider $N > 1$ controlled Markov chains $\{X_n^\alpha, n \geq 0\}$, $1 \leq \alpha \leq N$, on a finite state space $S = \{1, 2, \dots, d\}$, $1 < d < \infty$ with control space $\mathcal{U} := \{0, 1\}$. The controlled transition kernel

$$(i, j, u) \in S^2 \times \mathcal{U} \mapsto p^\alpha(j|i, u) \in [0, 1]$$

for the α -th chain satisfies $\sum_j p^\alpha(j|i, u) = 1 \forall i, u$, and has the interpretation of ‘probability of going from state i to state j under control u ’. The control variable u is binary, corresponding to two modes of operation, active ($u = 1$) and passive ($u = 0$). Define the increasing family of σ -fields $\mathcal{F}_n := \sigma(X_m^\alpha, U_m^\alpha, 1 \leq \alpha \leq N, m \leq n)$, $n \geq 0$. The ‘controlled Markov property’ is

$$P(X_{n+1}^\alpha = j | \mathcal{F}_n) = p^\alpha(j | X_n^\alpha, U_n^\alpha), \forall n \geq 0, 1 \leq \alpha \leq N,$$

where $\{U_n^\alpha\}_{n \geq 0}$, $1 \leq \alpha \leq N$, are the $\mathcal{U}$ -valued control processes, called ‘admissible controls’. A special subclass denoted SP is that of stationary policies wherein $U_n^\alpha = \varphi^\alpha(X_n^\alpha)$ for some $\varphi^\alpha : S \mapsto \{0, 1\}$, $n \geq 0$. The individual chains are called ‘arms’ of the restless bandit. Let $r^\alpha : (i, u) \in S \times \mathcal{U} \mapsto \mathcal{R}$ denote prescribed per stage reward function for the α -th chain. These controlled Markov chains are assumed to satisfy:

(C0) (Unichain property) There exists a distinguished state $i_0 \in S$ that is reachable with strictly positive probability from any other state under any stationary policy (SP).

Since $d = |S| < \infty$, this implies in particular that for $\tau := \min\{n \geq 0 : X_n^\alpha = i_0\}$,

$$\max_{k \in S, SP} E[\tau | X_0^\alpha = k] < \infty. \quad (1)$$

The objective is to maximize the long run average reward

$$\liminf_{n \uparrow \infty} \frac{1}{n} E \left[\sum_{m=0}^{n-1} \sum_{\alpha=1}^N r^{\alpha}(X_m^{\alpha}, U_m^{\alpha}) \right], \quad (2)$$

subject to the constraint: for a prescribed $M < N$,

$$\sum_{\alpha=1}^N U_n^{\alpha} = M, \quad \forall n. \quad (3)$$

That is, at each time instant, only M arms are activated.

That is, at each time instant, only M arms are activated.

The Whittle relaxation is to replace the ‘per time instant’ constraint (3) by a ‘time-averaged constraint’

$$\liminf_{n \uparrow \infty} \frac{1}{n} E \left[\sum_{m=0}^{n-1} \sum_{\alpha=1}^N U_m^{\alpha} \right] = M. \quad (4)$$

That is, we consider now the unconstrained control problem of maximizing

$$\liminf_{n \uparrow \infty} \frac{1}{n} E \left[\sum_{m=0}^{n-1} (r^\alpha(X_m^\alpha, U_m^\alpha) + \lambda(1 - U_m^\alpha)) \right] \quad (5)$$

separately for each α .² The dynamic programming equation then is

$$\begin{aligned} V^\alpha(i) &= \max \left(r^\alpha(i, 1) + \sum_j p^\alpha(j|i, 1) V^\alpha(j), \right. \\ &\quad \left. r^\alpha(i, 0) + \lambda + \sum_j p^\alpha(j|i, 0) V^\alpha(j) \right) - \beta^\alpha \\ &= \max_{u \in \mathcal{U}} \left(u(r^\alpha(i, 1) + \sum_j p^\alpha(j|i, 1) V^\alpha(j)) + (1 - u) \times \right. \\ &\quad \left. (r^\alpha(i, 0) + \lambda + \sum_j p^\alpha(j|i, 0) V^\alpha(j)) \right) - \beta^\alpha, \end{aligned} \quad (6)$$

with $(V^\alpha(\cdot), \beta^\alpha) \in \mathcal{R}^d \times \mathcal{R}$ the unknown variables. Under **(C0)**, β^α is unique and equals the optimal reward. V is unique up to an additive constant. The optimal decision $u^*(i)$ in state i then is given by the maximizer on the right hand side of (6) (Puterman, 1994). If we consider individual arms separately, the superscript α will be dropped henceforth, used only when it is needed. Note, however, that this does not mean that all arms are assumed statistically identical.

$$v^\alpha(i) := \max_{u \in \mathcal{U}} \left(u(r^\alpha(i, 1) + \sum_j p^\alpha(j|i, 1)V^\alpha(j)) + (1 - u) \times \right. \\ \left. (r^\alpha(i, 0) + \lambda + \sum_j p^\alpha(j|i, 0)V^\alpha(j)) \right) - \beta^\alpha, \quad (7)$$

Define the Q-value as

$$Q(i, u) := u(r(i, 1) + \sum_j p(j|i, 1)V(j)) + (1 - u) \times \\ (r(i, 0) + \lambda + \sum_j p(j|i, 0)V(j)) - \beta. \quad (8)$$

This satisfies the equation

$$Q(i, u) = ur(i, 1) + (1 - u)(\lambda + r(i, 0)) - \\ \beta + \sum_j p(j|i, u) \max_v Q(j, v), \quad (9)$$

for $i \in S, u \in U$. Under **(C0)**, this has a solution (Q, β) where β is uniquely specified as the optimal reward and Q is unique up to an additive scalar, just as for (6). The set $\{j \in S : u^*(j) = 1\}$ is the set of states when the arm is active, its complement being the set of states when it is passive. Whittle's insight was to view the Lagrange multiplier as a 'subsidy' for passivity. He defined the problem to be indexable when the set of passive states increases monotonically from the empty set to all of S as the subsidy is increased from $-\infty$ to ∞ . In this case, he defines the (Whittle) index for state $\hat{k}$ to be the value $\lambda(\hat{k})$ of λ for which both active and passive modes are equally preferred in state $\hat{k}$. That is,

$$Q(i, u) = ur(i, 1) + (1 - u)(\lambda + r(i, 0)) - \beta + \sum_j p(j|i, u) \max_v Q(j, v),$$

and passive modes are equally preferred in state $\hat{k}$. That is,

$$\begin{aligned} \lambda(\hat{k}) := & r(\hat{k}, 1) + \sum_j p(j|\hat{k}, 1)V(j) - r(\hat{k}, 0) \\ & - \sum_j p(j|\hat{k}, 0)V(j). \end{aligned} \tag{10}$$

This is equivalent to solving

$$Q(\hat{k}, 1) - Q(\hat{k}, 0) = 0, \tag{11}$$

for $\lambda = \lambda(\hat{k})$, where the λ -dependence of the left hand side is not rendered explicit as per our convention thus far.

rendered explicit as per our convention that λ_n .

Our algorithm is a two time scale iteration wherein the faster timescale performs Q-learning for a 'static' subsidy λ_n , the latter in reality changing on a slower time scale. Thus it tracks the Q-value corresponding to the slowly changing subsidy, which in turn is updated on a slower timescale by a simple tuning scheme suggested by (11). The Whittle index is a function of $\hat{k} \in S$, so for large state spaces, one may compute it for a suitably chosen subset of S and interpolate.

3. Q-learning for Whittle index

Q-learning is one of the oldest and most popular reinforcement learning scheme for approximate dynamic programming, due to Watkins (Watkins, 1989). Originally developed for infinite horizon discounted rewards, we shall be using a variant for average reward from Abounadi et al. (2001). For the controlled Markov chain $\{X_n^\alpha\}$ above with average reward (5), the ‘RVI Q-learning’ algorithm of (2.7) in Abounadi et al. (2001) is as follows (with a key difference we highlight later). Fix a stepsize sequence $\{a(n)\}$ satisfying $\sum_n a(n) = \infty$ and $\sum_n a(n)^2 < \infty$. For each $i \in S$, $u \in \mathcal{U}$, do:

$$\begin{aligned} Q_{n+1}(i, u) &= Q_n(i, u) + a(v(i, u, n))I\{X_n = i, U_n = u\} \\ &\quad \times \left((1-u)(r(i, 0) + \lambda) + ur(i, 1) + \right. \\ &\quad \left. \max_{v \in \mathcal{U}} Q_n(X_{n+1}, v) - f(Q_n) - Q_n(i, u) \right), \end{aligned} \quad (12)$$

where³

$$f(Q) = \frac{1}{2d} \sum_{i \in S} (Q(i, 0) + Q(i, 1)). \quad (13)$$

Here for $i \in S$, $u \in \mathcal{U}$,

$$v(i, u, n) = \sum_{m=0}^n I\{X_m = i, U_m = u\},$$

is the ‘local clock’ for the pair (i, u) counting the updates of the (i, u) -th component.

Our objective is to learn the Whittle index, i.e., the value $\lambda(\hat{k})$ of λ defined in (10), equivalently in (11), for which active and passive modes are equally desirable for a given $\hat{k} \in S$. Hence we also have an updating scheme for λ , leading to a coupled iteration for each $\hat{k} \in S$. The first component is the same as (12) except for the replacement of λ by the estimated Whittle index $\lambda_n(\hat{k})$. Thus for each $\hat{k} \in S$, we perform the iteration

$$\begin{aligned} Q_{n+1}(i, u; \hat{k}) &= Q_n(i, u; \hat{k}) + a(v(i, u, n)) \times \\ &\quad I\{X_n = i, U_n = u\} \left((1-u)(r(i, 0) + \lambda_n(\hat{k})) + ur(i, 1) \right. \\ &\quad \left. + \max_{v \in \mathcal{U}} Q_n(X_{n+1}, v; \hat{k}) - f(Q_n(\hat{k})) - Q_n(i, u; \hat{k}) \right) \end{aligned} \quad (14)$$

along with an update for learning the Whittle index $\lambda(\hat{k})$ for state $\hat{k}$ given by: with a prescribed stepsize sequence $\{b(n)\}$ satisfying $\sum_n b(n) = \infty$, $\sum_n b(n)^2 < \infty$ and $b(n) = o(a(n))$, do

$$\lambda_{n+1}(\hat{k}) = \lambda_n(\hat{k}) + b(n) \left(Q_n(\hat{k}, 1; \hat{k}) - Q_n(\hat{k}, 0; \hat{k}) \right). \quad (15)$$

We use the ‘hat notation’ $\hat{k}$ to emphasize that $\hat{k}$ is the $\hat{k}$ -th component of the Whittle index estimation evolving on the slow time scale.

Note that we need to estimate $Q(i, u; \hat{k})$ for each arm α . However, if some arms are statistically identical, we can take advantage of this and collect statistics simultaneously from statistically identical arms. For instance, in the case of homogeneous arms and shared memory architecture, we need to update only $2d^2 + d$ variables, whereas they would have been $(2d)^N$ with Q-learning applied directly without the Whittle scheme.

The control actions at time n are defined as follows: Let $0 < \epsilon < 1$ be prescribed. With probability $(1 - \epsilon)$, we sort arms in the decreasing order of the estimated Whittle indices $\lambda_n(X_n^\alpha)$, $1 \leq \alpha \leq N$, and render the top M arms active, the remaining arms are passive. Ties are broken according to some pre-specified convention. With probability ϵ , we render active M random arms, chosen uniformly and independently, the rest passive. This uniform randomization with probability ϵ is essential in order to ensure that all state-action pairs are sampled ‘frequently’, i.e.,

$$\liminf_{n \uparrow \infty} \frac{v(i, u, n)}{n+1} \geq \Delta \text{ a.s.} \quad (16)$$

for some $\Delta > 0$ and all i, u . This is because by the martingale law of large numbers,

$$\begin{aligned} \lim_{n \uparrow \infty} \frac{1}{n} \left(\sum_{m=0}^n (I\{X_{m+1} = i, U_{m+1} = u\} \right. \\ \left. - P(U_{m+1} = u|i)p(i|X_m, U_m)) \right) = 0 \text{ a.s.} \end{aligned}$$

and therefore the l.h.s. of (16) equals

$$\liminf_{n \uparrow \infty} \frac{\sum_{m=0}^n I\{X_{m+1} = i, U_{m+1} = u\}}{n}$$

$$\begin{aligned} &\geq \frac{\sum_{m=0}^n \epsilon p(i|X_m, U_m)}{n} \\ &\geq \epsilon \min\{p(k|j, v) : p(k|j, v) > 0, k, j \in S, v \in \mathcal{U}\} > 0. \end{aligned}$$

This ensures (16), i.e., adequate exploration, though at the expense of settling for near-optimality rather than optimality. The prospect of slowly decreasing ϵ with n has been explored in literature, see, e.g., Singh, Jaakkola, Littman, and Szepesvári (2000).

We used following stepsize sequences, which gave good performance in practice:

$$a(n) = \frac{C}{\lceil \frac{n}{500} \rceil}, \quad b(n) = \frac{C'}{1 + \lceil \frac{n \log n}{500} \rceil} I\{n \pmod N \equiv 0\}. \quad (17)$$

Define $h(Q, \lambda) = [[h(Q, \lambda)_{iu}]]_{i \in S, u \in \{0,1\}} : \mathcal{R}^{2d} \times \mathcal{R} \mapsto \mathcal{R}^{2d}$ by

$$\begin{aligned} h(Q, \lambda)_{iu} &:= (1 - u)(r(i, 0) + \lambda) + ur(i, 1) \\ &\quad + \sum_j p(j|i, u) \max_{v \in \{0,1\}} Q(j, v) - f(Q). \end{aligned} \quad (18)$$

Letting $Q(\hat{k})$ denote $Q(\cdot, \cdot; \hat{k})$ suitably vectorized, we also define $M_n(\hat{k}) := [[M_n(\hat{k})_{iu}]]_{i \in S, u \in \{0,1\}}$ by

$$\begin{aligned} M_{n+1}(\hat{k})_{iu} &:= (1 - u)(r(i, 0) + \lambda_n(\hat{k})) + ur(i, 1) + \\ &\max_{v \in \mathcal{U}} Q_n(X_{n+1}, v; \hat{k}) - f(Q_n(\hat{k})) - h(Q_n(\hat{k}), \lambda_n(\hat{k}))_{iu}. \end{aligned} \quad (19)$$

Then $\{M_n(\hat{k})\}$ are martingale difference sequences w.r.t. $\{\mathcal{F}_n\}$, i.e., they are adapted to $\{\mathcal{F}_n\}$ and satisfy $E[M_{n+1}(\hat{k})_{iu} | \mathcal{F}_n] = 0 \forall i, \hat{k}, u, n$. Rewrite (14) as

$$\begin{aligned} Q_{n+1}(i, u; \hat{k}) &= Q_n(i, u; \hat{k}) \\ &+ a(v(i, u, n)) I\{X_n = i, U_n = u\} \\ &\times (h(Q_n(\hat{k}), \lambda_n(\hat{k}))_{iu} - Q_n(i, u; \hat{k}) + M_{n+1}(\hat{k})_{iu}). \end{aligned} \quad (20)$$

In view of the fact $b(n) = o(a(n))$, the coupled iterates (20), (15) form a two time scale stochastic approximation algorithm in the sense of Borkar (2009), Section 6.1, with (20) operating on the faster time scale and (15) on the slower time scale. We exploit this fact later in the convergence analysis.

4.1. Example with circulant dynamics

We first test our scheme on the example from [Fu et al. \(2019\)](#). The example has four states and the dynamics are circulant: when an arm is passive ($u = 0$), resp. active ($u = 1$), the state evolves according to the transition probability matrices

$$P_0 = \begin{bmatrix} 1/2 & 0 & 0 & 1/2 \\ 1/2 & 1/2 & 0 & 0 \\ 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 1/2 & 1/2 \end{bmatrix}, \quad \text{and} \quad P_1 = P_0^T,$$

respectively. The rewards do not depend on the action and are given by $r(1, 0) = r(1, 1) = -1$, $r(2, 0) = r(2, 1) = 0$, $r(3, 0) = r(3, 1) = 0$, and $r(4, 0) = r(4, 1) = 1$. Intuitively, there is a preference to activate an arm when the arm is in state 3. Indeed, the exact values of the Whittle indices, calculated in [Fu et al. \(2019\)](#), are as follows: $\lambda(1) = -1/2$, $\lambda(2) = 1/2$, $\lambda(3) = 1$, and $\lambda(4) = -1$, which give priority to state 3. Consider a scenario with $N = 100$ arms, out of which $M = 20$ are active at each time. We initialize our algorithm with $\lambda_0(i) = 0$, and $Q(i, u) = r(i, u)$, $\forall i \in S$.

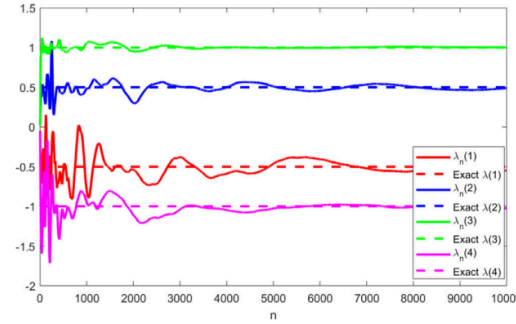


Fig. 1. Estimated (solid lines) and exact (dash lines) Whittle indices in the example with circulant dynamics.

In this example, we assumed the shared memory architecture and took full advantage of the fact that the arms are statistically identical. This helps to collect the statistics very quickly and results in a rapid convergence of the algorithm. We first set the exploration parameter as $\epsilon = 0.1$.

In [Fig. 1](#) we present the convergence of the estimated values of the Whittle indices (see (15)) to the exact values. In [Fig. 2](#), we present the comparison of the running average reward obtained by our algorithm with that of the algorithm based on the use of the exact Whittle indices from the beginning. We see that the average rewards stabilize in both approaches already after 250 iterations. The 10% loss of efficiency of our scheme with respect to the approach using the exact Whittle indices is due to the fact that we spend 10% of effort on pure exploration. This actually can be mitigated by decreasing the exploration parameter with time. We notice that as predicted by the theory and confirmed by [Fig. 1](#) the estimated Whittle indices in our algorithm converge to the true values.

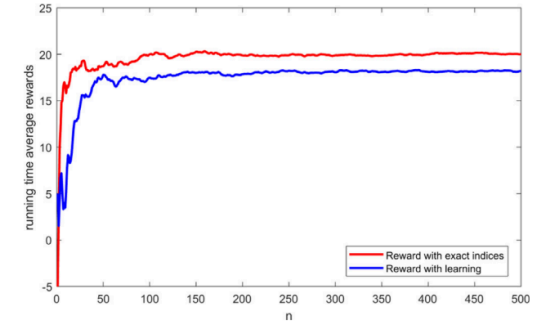


Fig. 2. Rewards comparison in the circulant dynamics ($\epsilon = 0.1$).

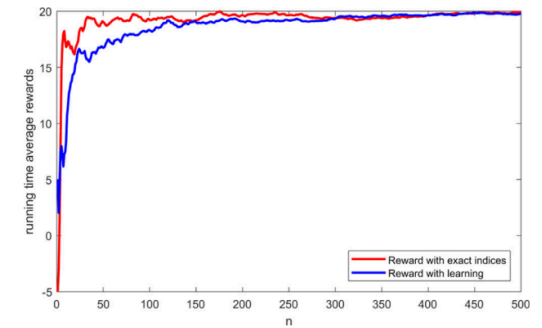


Fig. 3. Rewards comparison in the circulant dynamics ($\epsilon = 0.01$).

Lagrange Index...

Lagrange Index Policy for Restless Bandits with Average Reward

Konstantin Avrachenkov¹, Vivek S. Borkar² and Pratik Shah³

Abstract: We study the Lagrange Index Policy (LIP) for restless multi-armed bandits with long-run average reward. In particular, we compare the performance of LIP with the performance of the Whittle Index Policy (WIP), both heuristic policies known to be asymptotically optimal under certain natural conditions. Even though in most cases their performances are very similar, in the cases when WIP shows bad performance, LIP continues to perform very well. We then propose reinforcement learning algorithms, both tabular and NN-based, to obtain online learning schemes for LIP in the model-free setting. The proposed reinforcement learning schemes for LIP require significantly less memory than the analogous schemes for WIP. We calculate analytically the Lagrange index for the restart model, which applies to the optimal web crawling and the minimization of the weighted age of information. We also give a new proof of asymptotic optimality in case of homogeneous arms as the number of arms goes to infinity, based on exchangeability and de Finetti's theorem.

Key words: multi-armed bandits; restless bandits; Lagrange index; Whittle index; exchangeability

1.2 Model Formulation and Index Policies

Consider $N > 1$ controlled discrete-time Markov chains (*arms* in bandit terminology) $X_n^i, n \geq 0$, for $1 \leq i \leq N$, on finite state space $\mathcal{X}$, controlled by the $\mathcal{U} := \{0, 1\}$ -valued control sequences $U_n^i, n \geq 0$. Given the control U_n^i , the i -th chain moves from the current state $X_n^i \in \mathcal{X}$ to the next state $y \in \mathcal{X}$ according to the transition probability kernel $p^i(y|X_n^i, U_n^i)$ and obtains the reward $r^i(X_n^i, U_n^i)$. We seek to solve the following long-run average reward MDP problem:

(P_0) Maximize the long-run average reward

$$\lim_{n \rightarrow \infty} \frac{1}{n} E \left[\sum_{m=0}^{n-1} \sum_{i=1}^N r^i(X_m^i, U_m^i) \right], \quad (1.1)$$

subject to the constraint

$$\sum_{i=1}^N U_n^i = M, \quad \forall n \geq 0, \quad (1.2)$$

for a prescribed $1 \leq M < N$. The constraint (1.2) means that we need to set exactly M arms active at each time step.

We shall consider the Whittle relaxation [53] of this problem, where we replace (1.2) by an average constraint. Thus the problem becomes

(P_1) Maximize the average reward

$$\lim_{n \rightarrow \infty} \frac{1}{n} E \left[\sum_{m=0}^{n-1} \sum_{i=1}^N r^i(X_m^i, U_m^i) \right], \quad (1.3)$$

subject to the constraint

$$\lim_{n \rightarrow \infty} \frac{1}{n} E \left[\sum_{m=0}^{n-1} \sum_{i=1}^N I\{U_m^i = 1\} \right] = M, \quad (1.4)$$

for a prescribed $1 \leq M < N$.

(P_2) Maximize the reward

$$\lim_{n \rightarrow \infty} \frac{1}{n} E \left[\sum_{m=0}^{n-1} \sum_{i=1}^N \left(r^i(X_m^i, U_m^i) + \lambda_* \left(I\{U_m^i = 1\} - \frac{M}{N} \right) \right) \right],$$

where λ_* is the Lagrange multiplier.

This separates the original MDP on S^N into N individual identical MDPs on S that seek to maximize

$$\lim_{n \rightarrow \infty} \frac{1}{n} E \left[\sum_{m=0}^{n-1} \left(r^i(X_m^i, U_m^i) + \lambda_* \left(I\{U_m^i = 1\} - \frac{M}{N} \right) \right) \right], \quad i = 1, \dots, N. \quad (1.5)$$

This is possible because both the objective function and the constrained function are separable in the arms. We assume that these individual MDPs are irreducible under all stationary policies.

Define the ‘Q-values’ $Q^i(x, u)$, $i = 1, \dots, N$, $x \in S$, $u \in \mathcal{U}$, by

$$Q^i(x, u) = ur^i(x, 1) + (1 - u)(\lambda_* + r^i(x, 0)) + \sum_y p^i(y|x, u) \max_v Q^i(y, v).$$

Let us now define the *Lagrange index* for the chains as follows:

$$\gamma^i(x) := Q^i(x, 1) - Q^i(x, 0). \tag{1.6}$$

This is the same as the ‘LP-index’ of [26], because the LP formulation and the Lagrange formulation for (P_1) are equivalent. The Lagrange index also belongs to the set of *priority policies* defined in [49].

Compare this with the Whittle index which is based on a subsidy λ given for staying passive, and the index for state x is the value of subsidy that makes the active and passive modes equally desirable. In particular, the Q-values in this case depend on the subsidy λ . If we denote this dependence explicitly by writing $Q_\lambda^i(x, u)$ for the Q-values, then the Whittle index for state x is that value of $\lambda = \lambda(x)$ for which λ is a solution of $Q_\lambda^i(x, 1) - Q_\lambda^i(x, 0) = 0$. This is precisely the condition under which both control choices are equally preferred, because their Q-values coincide. This is a system of equations that needs to be solved for *each* x . This requires a separate iterative scheme for each state x unless some other method allows one to explicitly compute the Whittle indices directly. This is usually not possible or easy without some additional structural properties in the problem that facilitate this route.

In contrast, the Lagrange index for x would correspond to $\gamma^i(x) = Q_{\lambda^*}^i(x, 1) - Q_{\lambda^*}^i(x, 0)$. This makes the computation of Lagrange index much easier. Specifically, it needs a single scalar ‘dual’ iteration in the primal-dual scheme. Compare this with the situation for Whittle index, where, except for the cases such as [6] where an explicit expression for Whittle index is available (and these are relatively very few), one needs one such iteration for *each* state [7].

2 Learning Algorithms for Lagrange Index

2.1 Tabular Q-Learning for Lagrange Indices

Algorithm 1 RL for Lagrange index with relaxed constraint

- 1: **initialize** $\lambda, Q(x, u)$ for all states and actions, ϵ
- 2: **for** $n = 1: n_{end}$ **do**
- 3: Choose action U_n^i for each arm i in ϵ -greedy fashion according to (2.5)
- 4: Update X_{n+1}^i and reward r_n^i from X_n^i and U_n^i for every arm i
- 5: Update (X_n^i, U_n^i) -th Q-value for each arm i as in (2.2)
- 6: Update common subsidy for passivity λ as in (2.4)
- 7: Optional: decrement ϵ as $\epsilon_{n+1} = \max(0.01, \epsilon_n * 0.99)$
- 8: **end for**
- 9: Calculate the new estimate of the index for each arm i :

$$\gamma_{n+1}^i(X_n^i) = Q_n^i(X_n^i, 1) - Q_n^i(X_n^i, 0)$$

$$U_n^i = \begin{cases} \arg \max_u Q^i(X_n^i, u), & \text{w.p. } 1 - \epsilon, \\ \text{Uni}(\mathcal{U}), & \text{w.p. } \epsilon. \end{cases} \quad (2.5)$$

$$Q_{n+1}^i(x, u) = Q_n^i(x, u) + \alpha(\nu(x, u, n, i)) I\{X_n = x, Z_n = u\} \times \\ \left(ur(x, 1) + (1 - u)(\lambda_n + r(x, 0)) + \max_v Q_n^i(X_{n+1}, v) - f(Q_n^i) - Q_n^i(x, u) \right), \quad (2.2)$$

where $X_{n+1}^i \sim p^i(\cdot | x, u)$, $\nu(x, u, n, i)$ counts the number of times $Q^i(x, u)$ has been updated till time n , and where

$$f(Q) = \frac{1}{2|\mathcal{X}|} \sum_{i \in S} (Q(i, 0) + Q(i, 1)). \quad (2.3)$$

The Lagrange multiplier is estimated by the dual iterates on the slower time scale:

$$\lambda_{n+1} = \lambda_n - \beta(n) \left(\sum_{i=1}^N U_n^i - M \right), \quad (2.4)$$

The step sizes $\{\alpha(n)\}, \{\beta(n)\}$ need to satisfy the following conditions:

$$\sum_n \alpha(n) = \infty, \quad \sum_n \alpha(n)^2 < \infty,$$

$$\sum_n \beta(n) = \infty, \quad \sum_n \beta(n)^2 < \infty, \quad \beta(n) = o(\alpha(n)). \quad (2.6)$$

2.2 Deep Q-Learning for Lagrange Indices

Algorithm 3 DQN Algorithm for Lagrange Index

```

1: initialize  $\lambda^*$ , DQN policy, DQN target, Batch Size,  $\epsilon=1$ 
2: for  $n = 1: n_{end}$  do
3:   Update  $\beta(n)$  as:  $\beta(n) = \frac{1}{\lceil n \log(n)/5000 \rceil + 1}$ 
4:   for arm  $i \in N$  do
5:     Choose action  $U_n^i$  for each arm  $i$  in an  $\epsilon$ -greedy fashion
6:     Update  $X_{n+1}^i$  and reward  $r_n^i$  from  $X_n^i$  and  $U_n^i$  for every arm  $i$ 
7:     Store  $(X_n^i, U_n^i, r_n^i, X_{n+1}^i, \lambda_n)$  in experience replay buffer
8:     Train DQN policy model on one batch of  $(X_n^i, U_n^i, r_n^i, X_{n+1}^i, \lambda_n)$  tuples from experience replay buffer
9:   end for
10:  Update the DQN target model
11:  Update common subsidy for passivity for all arms  $\lambda$  as:
12:   $\lambda_{n+1} = \lambda_n + \beta(n) \left( \sum_{i=1}^{i=N} U_n^i - M \right)$ 
13:  Decrement  $\epsilon$  as  $\epsilon_{n+1} = \max(0.01, \epsilon_n * 0.99)$ 
14: end for
15: Calculate the new index for every state  $s$  of each arm  $i$  by:
     $\gamma^i(x) = Q_\theta(x, 1) - Q_\theta(x, 0)$ 

```

The equation for the Q target used in training DQN is:

$$Q_{target}(X_n, U_n) \leftarrow ((1 - a_n)(r_0(X_n) + \lambda_n) + a_n r_1(X_n) + \max_{v \in \{0,1\}} Q_n(X_{n+1}, v) - \frac{1}{2d} \sum_{k \in S} (Q_n(k, 0) + Q_n(k, 1))). \quad (2.8)$$

Note that in DQN we employ a second neural network, called the target model, for the computation of $\max_{v \in \{0,1\}} Q_n(X_{n+1}, v)$. The reason for this decision is due to the maximisation bias of Q-learning: overestimating the Q-values results in the error increasing over time, due to the target being $r + \gamma \max_a Q(i, a)$. The use of a second neural network for the target helps control this bias. This second neural network copies periodically the parameter values of the main network.

We store the tuples $(X_n^i, U_n^i, r_n^i, X_{n+1}^i, \lambda_n)$ in an experience replay buffer, from which a batch of random samples is taken every time to train the main neural network. For each tuple $(X_n^i, U_n^i, r_n^i, X_{n+1}^i, \lambda_n)$ we calculate Q target values using equation (2.8). Once we have calculated Q-values for all the examples in batch through $Q(\cdot)$ and its targets $Q_{target}(\cdot)$, we compute the loss function as the mean square error between the two and train the main neural network, using a standard Adam optimiser.

- [54] Yu, Z., Xu, Y. and Tong, L., 2018. Deadline scheduling as restless bandits. IEEE Trans. on Automatic Control, 63(8), 2343-2358.

Deadline scheduling as restless bandits
 Z.Yu, Y.Xu, L.Tong
 IEEE Transactions on Automatic Control, 2018 · [ieeexplore.ieee.org](#)

The problem of stochastic deadline scheduling is considered. A constrained Markov decision process model is introduced in which jobs arrive randomly at a service center with stochastic job sizes, rewards, and completion deadlines. The service provider faces random processing costs, convex noncompletion penalties, and a capacity constraint that limits the simultaneous processing of jobs. Formulated as a restless multiarmed bandit problem, the stochastic deadline scheduling problem is shown to be indexable. A closed-

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3.3 Deadline Scheduling Problem

Let us now consider the deadline scheduling problem studied in [54]. The states of this problem are formed by two different variables: the service time $B \in [0, 9]$ and the deadline $T \in [0, 12]$, leading to a total of $|S| = 130$ states (although several of these states are not accessible by the underlying Markov chain). The variable B represents the amount of workload pending to complete a certain job while the variable T represents the remaining time available to perform it. When there is no job at the i^{th} position, the state is $(0, 0)$, while otherwise it is (T_n^i, B_n^i) . In each iteration, the state transition from $X_n^i = (T_n^i, B_n^i)$ depends on the action U_n^i in the following way:

$$s_{n+1}^i = \begin{cases} (T_n^i - 1, (B_n^i - U_n^i)^+), & \text{if } T_n^i > 1, \\ (T, B) \text{ with prob. } q(T, B), & \text{if } T_n^i \leq 1. \end{cases} \quad (3.1)$$

where $b^+ = \max(b, 0)$. When $T = 1$, the deadline for completing the job ends, and the new state (including the empty state $(0, 0)$) is chosen uniformly at random. The value of B , the workload to be completed, is only reduced if the action is active. If the scheduler reaches the state $(T = 1, B > 0)$, the job cannot be finished on time, and an extra penalty $F(B_n^i - a_n^i) = 0.2(B_n^i - a_n^i)^2$ is incurred. In addition, activating the arms involves a fixed cost $c = 0.8$. The rewards are given by

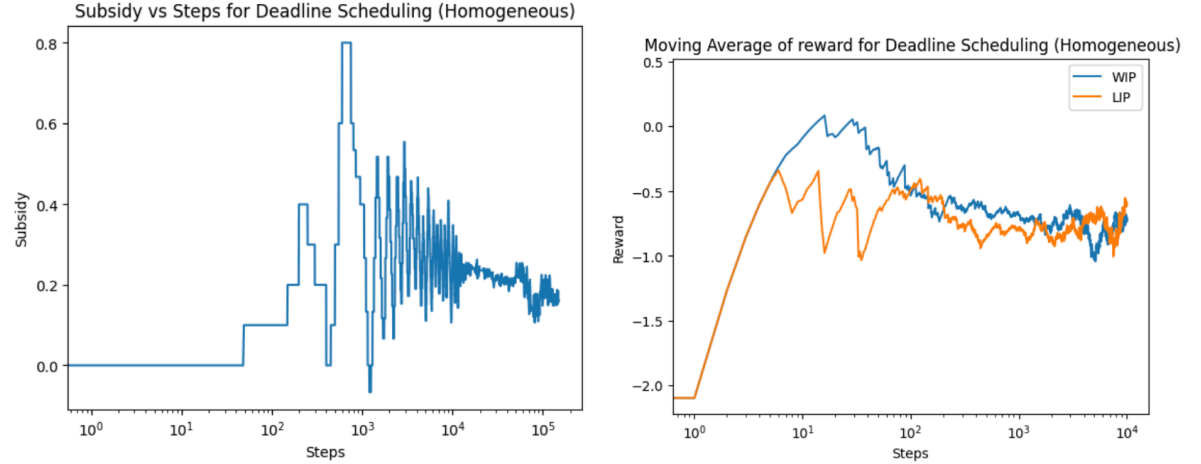
$$r_n^i(X_n^i, U_n^i, c) = \begin{cases} (1 - c)U_n^i, & \text{if } B_n^i > 0 \text{ and } T_n^i > 1, \\ (1 - c)U_n^i - F(B_n^i - a_n^i), & \text{if } B_n^i > 0 \text{ and } T_n^i = 1, \\ 0, & \text{otherwise.} \end{cases} \quad (3.2)$$

The authors of [54] provide an explicit expression for Whittle index, given by:

$$\lambda(T, B, c) = \begin{cases} 0, & \text{if } B = 0, \\ 1 - c, & \text{if } 1 \leq B \leq T - 1, \\ F(B - T - 1) - F(B - T) + 1 - c, & \text{if } T \leq B. \end{cases} \quad (3.3)$$

As the state space is large we use DQN to learn the Lagrange indices. A neural network with 3 hidden layers having (512, 256, 128) neurons connected through the ReLU activation is used. A batch size of 32 is used and the size of the experience replay buffer is 1000. The exploration decay factor is 0.9995 and the learning rate for the Adam optimiser is 10^{-5} .

We first consider the homogeneous case using $N = 5$ and $M = 2$, with a processing cost $c = 0.8$. The results are presented in Figure 5.

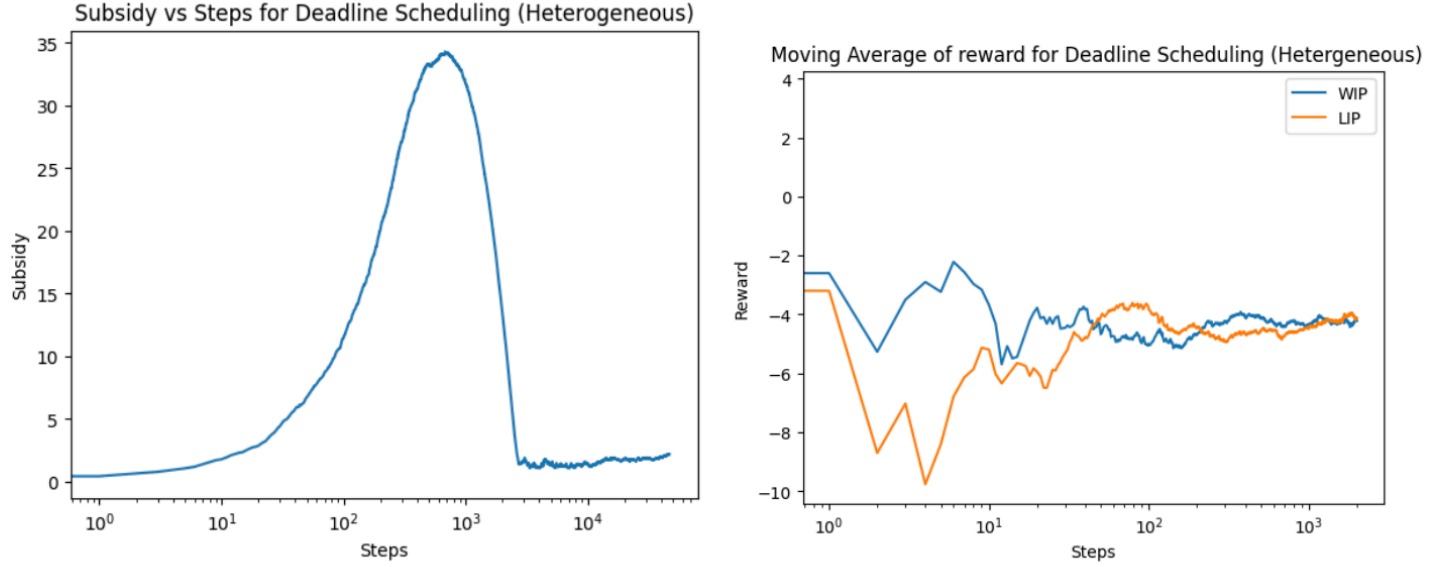


(a) Subsidy (Lagrange multiplier) estimation.

(b) Moving average reward.

Figure 5: Deadline scheduling problem (homogeneous arms). Algorithm 3.

Then we consider the scenario of heterogeneous arms, where each arm has a state space of size $|S| = 130$ and $N = 20$. The results are presented in Figure 6. To differentiate the arms, we assign different activation cost values $c = \{0.1, 0.3, 0.6, 0.8\}$ for every 5 arms. As was suggested in Section 2.2, we take into account heterogeneity by increasing the dimension of the input to the neural network. The first dimension is used to give information about the type of arm and the other dimensions represent the state of the arm. Specifically, the values of the first input are 0, 1, 2, 3 for $c = 0.1$, $c = 0.3$, $c = 0.6$ and $c = 0.8$.



(a) Subsidy (Lagrange multiplier) estimation.

(b) Moving average reward.

Figure 6: Deadline scheduling problem (heterogeneous arms). Algorithm 3.